

Highest Capacity Arc Lamps

Research Report | Generated June 24, 2026

This report identifies the six highest-wattage, highest-amperage, and highest-voltage arc lamps commercially available or assemblable from parts as of June 2026. All products are sourced from open web research across manufacturer, distributor, and wholesale channels. Each entry includes electrical specifications, application context, and direct purchase links.

Summary Ranking

Rank	Model	Wattage	Voltage (DC)	Current (DC)	Type	Cooling
1	Ushio UXW-15KD	15,000 W	~85–100 V	~150 A	Xe Short Arc	Water
2	ASL HVF12000CL/R	12,000 W	87–101 V	93–110 A	Xe Short Arc	Air/Water
3	SQP / ASL 10kW	10,000 W	~100–130 V	~80–100 A	Xe Short Arc	Air/Water
4	Osram XSTAGE 7000W OFR	7,000 W	~60–80 V	~90–115 A	Xe High-Pressure	Air
5	Philips LTI LTIX-7000W-HS	7,000 W	~60–80 V	~90–115 A	Xe Short Arc	Air
6	Hilltech Hg-Xe 5000W	5,000 W	~60–80 V	~65–80 A	Hg-Xe Short Arc	Air

#1 — Ushio UXW-15KD

15,000 W — Water-Cooled Xenon Short Arc (IMAX-Grade)

Power	15,000 W (15 kW)
Type	Water-cooled xenon short arc
Voltage	~85–100 V DC (operating)
Current	~150 A DC (operating)
Rated Life	1,000 hours
Color Temp	~6,000 K (daylight)
Applications	IMAX theatres, solar simulators, giant-screen projection
Feature	Up to 30% more screen brightness vs. competitive lamps

Where to Buy / Sources:

- Ushio America — Official Product Page — <https://www.ushio.com/product/uxw-15kd-water-cooled-xenon-for-imax-theaters/>
- eBay Listing — <https://www.ebay.com/itm/164826307713>
- AV-iQ Distributor — <https://www.av-iq.com/avcat/ctl1642/index.cfm?manufacturer=ushio-america&product;=5000640>
- RnB Enterprises Distributor — <https://catalog.rnbenterprises.com/avcat/ctl977/index.cfm?manufacturer=ushio-america&product;=5000640>

#2 — ASL HVF12000CL/R

12,000 W — Xenon Short Arc Cinema Lamp

Power	12,000 W (12 kW)
Type	Xenon short arc
Voltage	87–101 V DC (rated operating range)
Current	93–110 A DC (rated operating range)
Burn Position	Vertical, anode up
Applications	Large-format cinema projection, studio lighting

Where to Buy / Sources:

- ASL Product Spec Page — <https://www.aslamps.com/lamp.php?id=HVF12000CL/R>
- ASL Full Products Catalogue — <https://www.aslamps.com/products.php>

#3 — SQP / ASL 10,000 W Xenon

10,000 W — Xenon Short Arc (Air or Water Cooled)

Power	10,000 W (10 kW)
Type	Xenon short arc, air-cooled or water-cooled variants
Voltage	~100–130 V DC (operating, varies by housing)
Current	~80–100 A DC
Applications	Graphic arts, photochemistry, medical, military, fiber optics
Notes	Available as lamp-only or as complete system with housing & PSU

Where to Buy / Sources:

- Superior Quartz Products (SQP) — Air-Cooled Lamps — <https://sqpuv.com/acsalamps.php>
- ASL Products — <https://www.aslamps.com/products.php>
- Alibaba — 10kW Xenon Lamp (from ~\$20.80/unit) — <https://www.alibaba.com/showroom/xenon-lamp-10000w.html>
- Lumina Power — Arc Lamp PSUs (up to 6.5 kW xenon) — <https://luminapower.com/arc-lamp-power-supplies/>

#4 — Osram XSTAGE 7000W OFR

7,000 W — Xenon High-Pressure (Ozone-Free, Outdoor)

Part No.	Osram 69485
Power	7,000 W (7 kW)
Type	Ozone-free (OFR) xenon high-pressure lamp
Applications	Outdoor event lighting, architectural illumination
Notes	Optimised for outdoor, high-intensity beam work

Where to Buy / Sources:

- ProLampSales — Osram XSTAGE 7000W OFR —

<https://www.prolampsales.com/products/osram-69485-xstage-7000w-ofr-7000-watts-xenon-high-pressure-lamp>

#5 — Philips LTI LTIX-7000W-HS

7,000 W — Xenon Short Arc (Standard Cinema)

Power	7,000 W (7 kW)
Type	Standard xenon short arc
Color Temp	~6,000 K (daylight)
Feature	High-intensity white light for film projection
Applications	Film & cinema projection

Where to Buy / Sources:

- ProLampSales — Philips LTIX-7000W-HS — <https://www.prolampsales.com/products/philips-lti-ltix-7000w-hs-7000w-standard-xenon-lamp>

#6 — Hilltech Mercury-Xenon 5,000 W

5,000 W — Mercury-Xenon Short Arc (UV/Research)

Power	5,000 W (5 kW) — top of 200–5,000 W product range
Type	Mercury-xenon (Hg-Xe) short arc
Voltage	~60–80 V DC (operating)
Current	~65–80 A DC
Applications	UV curing, photochemistry, research, optical instrumentation
Assembly	Lamp + housing + PSU purchased separately

Where to Buy / Sources:

- Hilltech — Xenon DC Short Arc Lamps — <https://www.hilltech.com/shrtarc.html>
- arc-lamps.com — Mercury-Xenon Manufacturer — http://www.arc-lamps.com/mercury_xenon_lamps.html
- Lumina Power — PSU (up to 5 kW mercury, 6.5 kW xenon) — <https://luminapower.com/arc-lamp-power-supplies/>

Notable Mention — Off-the-Charts Record

Vortek Water-Wall Plasma Arc Lamp — 300,000 W (300 kW)

Guinness World Record holder (1986 & 1993) for the most powerful continuously burning light source, producing over 1.2 million candlepower. Invented 1975. Not commercially purchasable; research/lab-built only. Operated via water-wall plasma confinement at extreme current densities.

Procurement Guide: Complete Product vs. Parts to Assemble

Option	What to Buy	Key Suppliers
Complete System (lamp + housing + PSU)	Plug-and-play light source systems	Newport/MKS, Thorlabs, Sciencetech Inc., Ushio kits
Lamp Only	Drop-in replacement or new build	ASL, SQP, Hilltech, Ushio, Osram, Philips via ProLampSales
PSU Only	Power supply for existing lamp housing	Lumina Power (650 W–6.5 kW), GlobalSpec directory
Bulk / Low-Cost	Xenon lamp bodies at volume price	Alibaba (from ~\$20.80/unit, MOQ 2)

Next research cycle: This report will be updated daily for 5 days, then monthly, targeting industrial-scale arc lamp systems rated at 250,000 W / 250,000 A / 250,000 V for sustained daily use (plasma arc furnaces, pulsed-power systems, high-energy research).

Sources: [ushio.com](https://www.usnio.com) | [aslamps.com](https://www.aslamps.com) | [sqpuv.com](https://www.sqpuv.com) | [hilltech.com](https://www.hilltech.com) | [luminapower.com](https://www.luminapower.com) | [newport.com](https://www.newport.com) | [prolampsales.com](https://www.prolampsales.com) | [alibaba.com](https://www.alibaba.com) | [globalspec.com](https://www.globalspec.com) | [verifiedmarketreports.com](https://www.verifiedmarketreports.com)